1. Order Form

Original table (UNF): one order with a repeating part group

customerNo	customerName	customerType	orderDate	orderTime	employeeName	partNo	partName	partType	partCode	quantityOrdered	unitPrice
HG54587	Jeff Peterson	Consumer	7/1/2024	10:30 am	D. Harrison	10654	Float Control	Plumbing	G413	4	12
						10456	Modulator	Electrical	H433	3	7
						10776	House Assembly	Plumbing	G413	7	9
						10657	Float Assembly	Plumbing	G413	5	10

1NF

order_line_flat

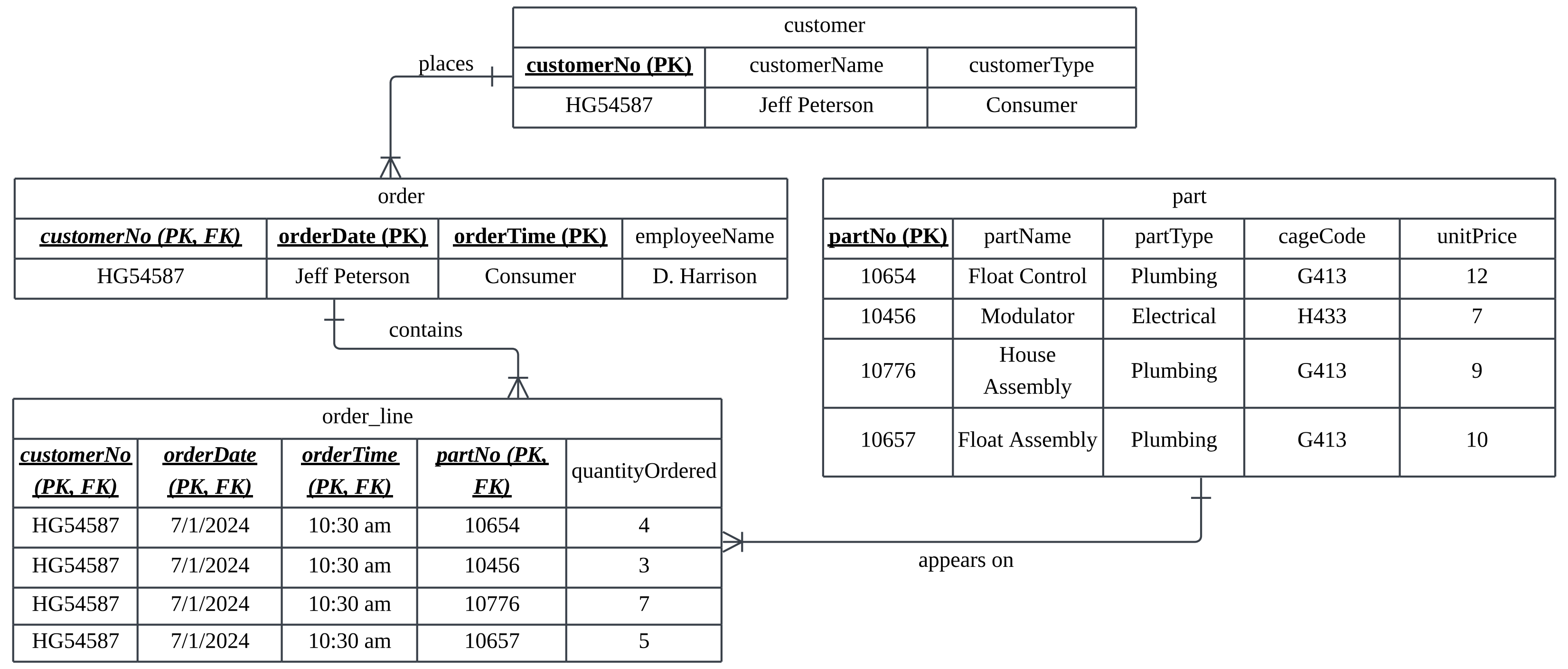
customerNo	customerName	customerType	**orderDate**	**orderTime**	employeeName	**partNo**	partName	partType	cageCode	quantityOrdered	unitPrice
HG54587	Jeff Peterson	Consumer	7/1/2024	10:30 am	D. Harrison	10654	Float Control	Plumbing	G413	4	12
HG54587	Jeff Peterson	Consumer	7/1/2024	10:30 am	D. Harrison	10456	Modulator	Electrical	H433	3	7
HG54587	Jeff Peterson	Consumer	7/1/2024	10:30 am	D. Harrison	10776	House Assembly	Plumbing	G413	7	9
HG54587	Jeff Peterson	Consumer	7/1/2024	10:30 am	D. Harrison	10657	Float Assembly	Plumbing	G413	5	10

UNF to 1NF: expand the repeating group to one row per ordered part and repeat the order header. Every cell is now atomic.

PK = (customerNo, orderDate, orderTime, partNo). Quantity depends on this entire key; the customer, order and part descriptions do not.

Functional dependencies:
customerNo -> customerName, customerType
(customerNo, orderDate, orderTime) -> employeeName
partNo -> partName, partType, cageCode, unitPrice
(customerNo, orderDate, orderTime, partNo) -> quantityOrdered

2NF + 3NF



customer

customerNo (PK)	customerName	customerType
HG54587	Jeff Peterson	Consumer

order

customerNo (PK, FK)	**orderDate (PK)**	**orderTime (PK)**	employeeName
HG54587	Jeff Peterson	Consumer	D. Harrison

part

partNo (PK)	partName	partType	cageCode	unitPrice
10654	Float Control	Plumbing	G413	12
10456	Modulator	Electrical	H433	7
10776	House Assembly	Plumbing	G413	9
10657	Float Assembly	Plumbing	G413	10

order_line

customerNo (PK, FK)	***orderDate (PK, FK)***	***orderTime (PK, FK)***	***partNo (PK, FK)***	quantityOrdered
HG54587	7/1/2024	10:30 am	10654	4
HG54587	7/1/2024	10:30 am	10456	3
HG54587	7/1/2024	10:30 am	10776	7
HG54587	7/1/2024	10:30 am	10657	5

1NF to 2NF: remove each partial dependency into customer, order and part. Keep quantityOrdered in orderLine because it depends on the complete line key. Customer data is separated now because customerNo is a proper subset of the original key.

2NF to 3NF: no further decomposition is needed. In every 2NF relation, non-key attributes depend on the key without a non-key intermediary.

Assumptions: (1) A customer places at most one order at a particular date and recorded time, so (customerNo, orderDate, orderTime) identifies an order.
(2) Each order has one assisting employee, and a part appears at most once per order.
(3) customerNo identifies one current customer name and type.
(4) partNo identifies one name, type, cage and fixed catalog unit price in this exercise. Prices are not order-specific.
(5) Each part occupies one cage; a cage may hold several part types.

2. Therapist Appointments

Original table (1NF)

therapist_appointment_flat

staffNo	therapistName	patNo	patName	**appointmentDate**	**appointmentTime**	branchNo
S1011	Fred Smith	P100	Lily White	9/12/2022	10:00	M15
S1011	Fred Smith	P105	Jill Baker	9/12/2022	12:00	M15
S1024	Heidi Pierce	P108	Andy McKee	9/12/2022	10:00	Q10
S1024	Heidi Pierce	P108	Andy McKee	9/14/2022	14:00	Q10
S1032	Richard Levin	P105	Jill Baker	9/14/2022	16:30	M15
S1032	Richard Levin	P110	Jimmy Winter	9/15/2022	18:00	B13

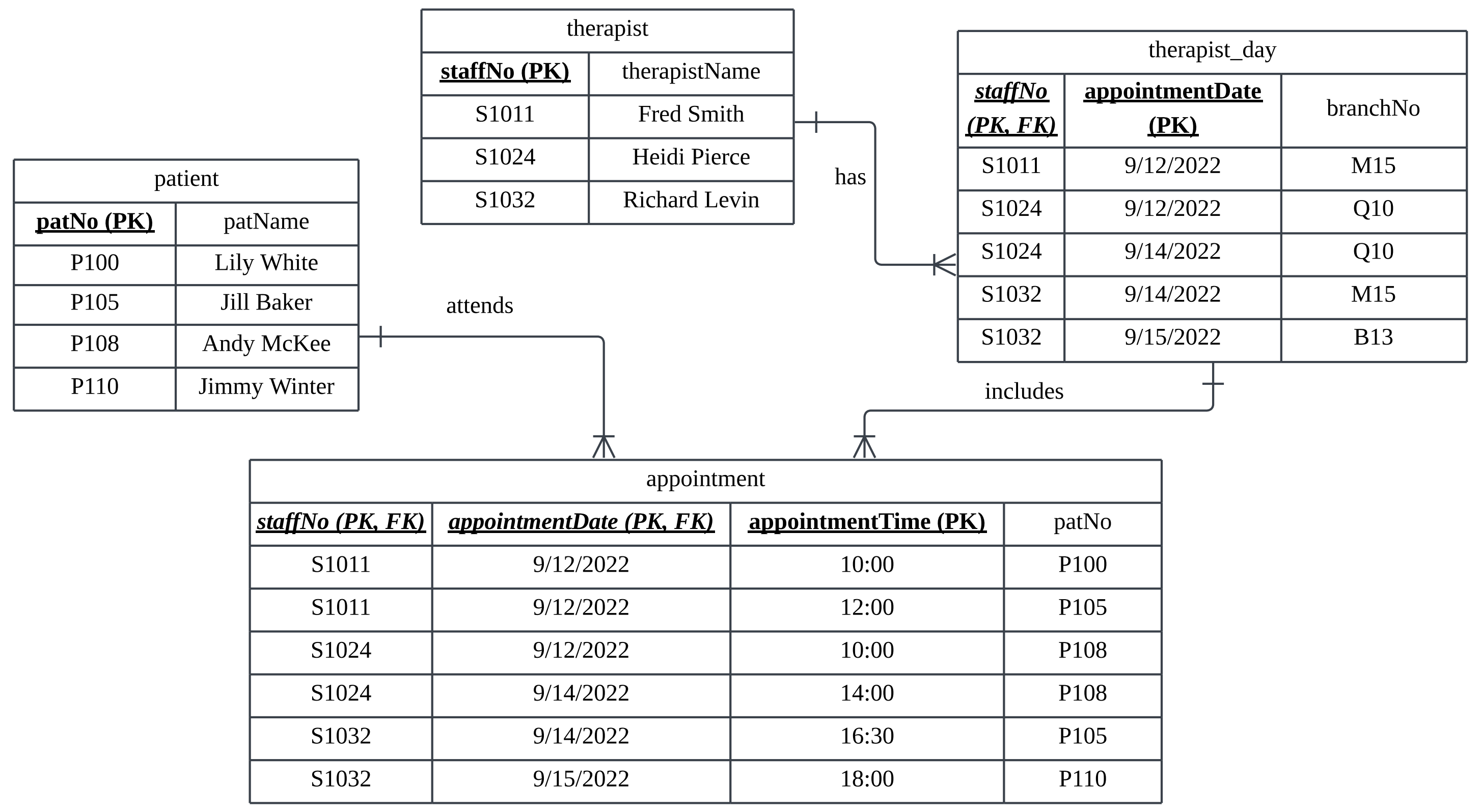
The source already shows one appointment per row, with separate date and time values under a shared appointment heading. Naming those components appointmentDate and appointmentTime makes the atomic attributes explicit. There is no repeating group, so the starting table is already in 1NF.

Chosen PK: (staffNo, appointmentDate, appointmentTime). Under the no-simultaneous-patient-appointments assumption, (patNo, appointmentDate, appointmentTime) is an alternate candidate key. Neither staffNo alone nor (staffNo, appointmentDate) identifies an appointment: S1011 has two appointments on 9/12/2022.

Functional dependencies:
staffNo -> therapistName
patNo -> patName
(staffNo, appointmentDate) -> branchNo
(staffNo, appointmentDate, appointmentTime) -> patNo
(patNo, appointmentDate, appointmentTime) -> staffNo (alternate-key assumption)

The table is not 2NF: therapistName depends on staffNo alone, and branchNo depends on staffNo plus appointmentDate. patName also depends on patNo, a proper subset of the alternate candidate key.

2NF + 3NF



therapist

staffNo (PK)	therapistName
S1011	Fred Smith
S1024	Heidi Pierce
S1032	Richard Levin

therapist_day

staffNo (PK, FK)	**appointmentDate (PK)**	branchNo
S1011	9/12/2022	M15
S1024	9/12/2022	Q10
S1024	9/14/2022	Q10
S1032	9/14/2022	M15
S1032	9/15/2022	B13

patient

patNo (PK)	patName
P100	Lily White
P105	Jill Baker
P108	Andy McKee
P110	Jimmy Winter

appointment

staffNo (PK, FK)	***appointmentDate (PK, FK)***	**appointmentTime (PK)**	patNo
S1011	9/12/2022	10:00	P100
S1011	9/12/2022	12:00	P105
S1024	9/12/2022	10:00	P108
S1024	9/14/2022	14:00	P108
S1032	9/14/2022	16:30	P105
S1032	9/15/2022	18:00	P110

1NF to 2NF: move therapistName to therapist, patName to patient, and the daily branch assignment to therapistDay. This removes partial dependencies on both candidate keys. appointment keeps the patient assigned to each therapist date/time slot. branchNo is not placed in therapist because staffNo alone does not determine it. It is not repeated in appointment because its determinant omits appointmentTime. Keeping one therapistDay row per therapist/date enforces the stated one-branch-per-day rule.

2NF to 3NF: the 2NF decomposition already has no transitive non-key dependencies, so the relations remain unchanged. Patient names and therapist names are each stored only with their own identifiers.

Assumptions: staffNo uniquely identifies a therapist and patNo uniquely identifies a patient. A therapist has at most one appointment at a given date and start time; appointments are individual sessions. A patient also cannot have two appointments at the same date and start time, but may have multiple appointments at different times that day. Names are not identifiers. A therapist works at exactly one branch on a day with appointments; the branch can change between days.

3. Event Staff

Original table (1NF)

event_staff_flat

eNo	**contractNo**	hours	eName	eventNo	eventLoc
1135	C1024	16	Smith J	H25	Queens
1057	C1024	24	Hocine D	H25	Queens
1068	C1025	28	White T	H4	Yonkers
1135	C1025	15	Smith J	H4	Yonkers
1135	C1026	10	Smith J	H25	Queens

PK = (eNo, contractNo). Neither component alone is unique: employee 1135 appears on three contracts, and contract C1024 includes two employees. All values are atomic and each row represents one employee/contract total, so no UNF-to-1NF restructuring is needed.

Functional dependencies:
eNo -> eName
contractNo -> eventNo
eventNo -> eventLoc
(eNo, contractNo) -> hours
Consequently, contractNo -> eventLoc also holds.

The table is not 2NF: eName depends on only eNo, while eventNo and eventLoc depend on only contractNo. It also contains the transitive dependency contractNo -> eventNo -> eventLoc. The H25 rows demonstrate that different contracts may refer to the same event.

2NF

employee

eNo (PK)	eName
1135	Smith J
1057	Hocine D
1068	White T

employee_contract

eNo (PK)	**contractNo (PK)**	hours
1135	C1024	16
1057	C1024	24
1068	C1025	28
1135	C1025	15
1135	C1026	10

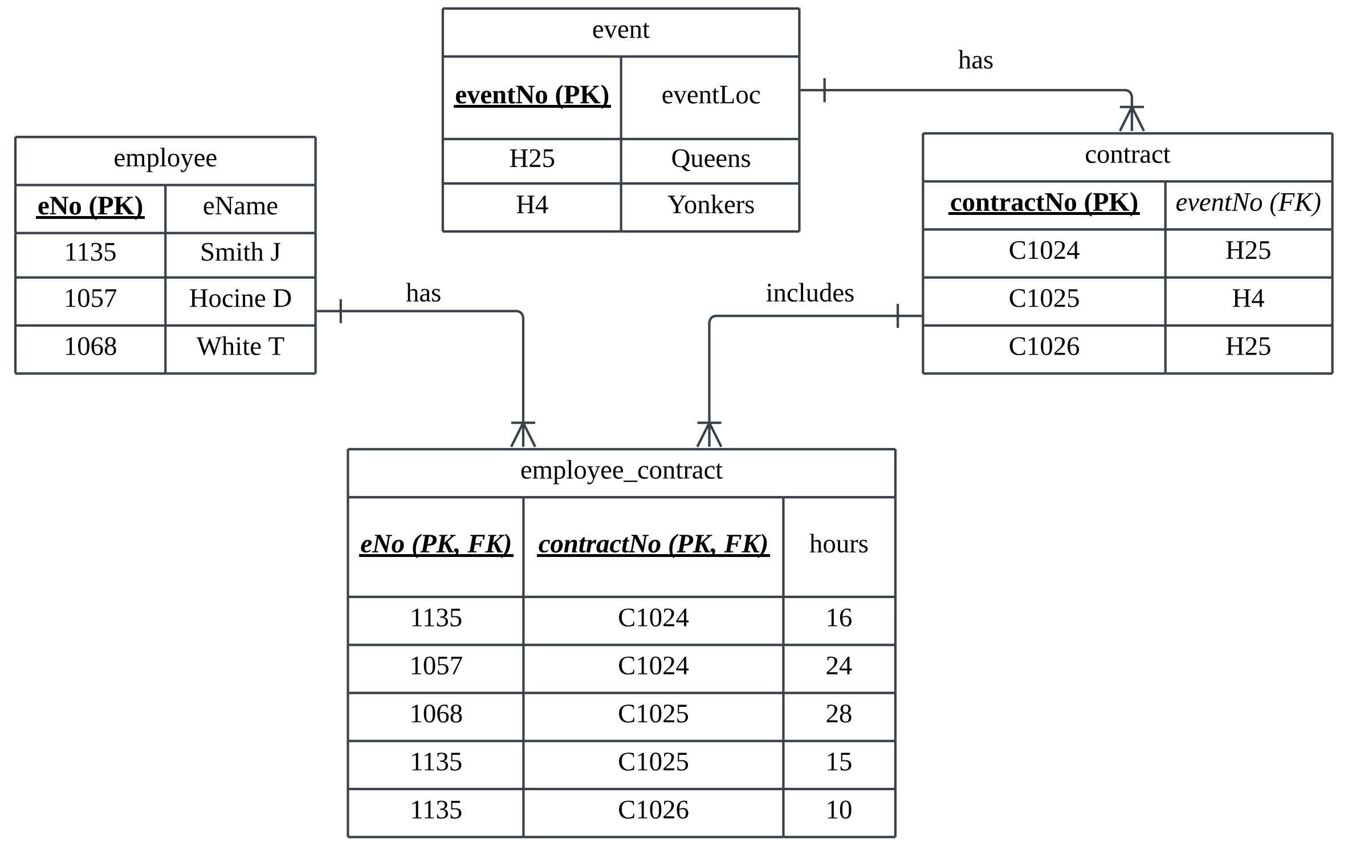
contract

contractNo (PK)	eventNo	eventLoc
C1024	H25	Queens
C1025	H4	Yonkers
C1026	H25	Queens

1NF to 2NF: separate employee facts by eNo and contract facts by contractNo. Leave hours in employeeContract because its determinant is the full composite key. Keep eventLoc with contract at this stage so that the remaining transitive dependency is visible. employee and contract have single-attribute primary keys, so they cannot have partial dependencies. In employeeContract, hours depends on both key components. These relations therefore satisfy 2NF.

contract is not yet in 3NF: contractNo determines eventNo, and the non-key attribute eventNo determines eventLoc. Queens is repeated for event H25 under C1024 and C1026; changing an event location would require multiple updates.

3NF



event

eventNo (PK)	eventLoc
H25	Queens
H4	Yonkers

employee

eNo (PK)	eName
1135	Smith J
1057	Hocine D
1068	White T

contract

contractNo (PK)	*eventNo (FK)*
C1024	H25
C1025	H4
C1026	H25

employee_contract

eNo (PK, FK)	***contractNo (PK, FK)***	hours
1135	C1024	16
1057	C1024	24
1068	C1025	28
1135	C1025	15
1135	C1026	10

2NF to 3NF: move eventLoc to event, keyed by eventNo. Retain eventNo in contract as the FK linking each contract to its single event. employee and employeeContract are unchanged. Join employeeContract to employee on eNo, to contract on contractNo, and then to event on eventNo. This reproduces all five source rows. Employee names, contract/event assignments, event locations and employee/contract hours can each be maintained in one appropriate relation.

Assumptions: eNo uniquely identifies an employee, contractNo uniquely identifies a contract, and eventNo uniquely identifies an event. Each event has one location. Each employee/contract pair occurs once; hours is the total number of hours for that employee on that contract, not a separate shift entry. An employee can work on several contracts and a contract can include several employees. Each contract covers exactly one event, while an event can have several contracts.